

INTRODUCTION TO COMPUTING

Assignment 3: Digital Logic Design

Instructions: For each of the scenarios below, read the description carefully to determine the logical conditions. Complete Tasks A, B, and C for every question. For Task C, use standard digital logic gates (AND, OR, NOT) and ensure your layout and notation follow the conventions established in the Princeton TOY 8 architectural design documentation.

Question 1: Environmental Warning Light

An industrial warehouse uses an environmental monitoring system that controls a red warning light. The system reads a temperature sensor and a pressure sensor. The temperature sensor outputs a 0 if the temperature is $<30^{\circ}\text{C}$, and a 1 if it is 30°C or higher. The pressure sensor outputs a 0 if the pressure is <50 psi, and a 1 if it is 50 psi or higher. There is also a master alarm switch (outputs 1 if activated). The warning light must turn on if the master switch is activated. If the master switch is off, the warning light should only turn on if the temperature is 30°C or higher AND the pressure is 50 psi or higher.

- **Task A:** Infer the logic from the description and create a complete truth table for the warning light based on the three inputs.
- **Task B:** Write out the Boolean logic expression for the warning light (simplification is optional).
- **Task C:** Draw the logic circuit diagram for this system.

Question 2: Automatic Sump Pump

A basement is equipped with an automatic sump pump to prevent flooding. The system uses a water level sensor that outputs a 1 if the water level is >10 cm, and 0 if it is ≤ 10 cm. A second sensor checks the weather forecast, outputting a 1 if heavy rain is expected. A backup battery sensor outputs a 1 if fully charged and 0 if it is $<20\%$ charged. The pump will activate (output 1) if the water level is >10 cm, regardless of other conditions. It will also activate preemptively if rain is expected AND the backup battery is fully charged.

- **Task A:** Infer the logic from the description and create a complete truth table for the sump pump.
- **Task B:** Write out the Boolean logic expression for the pump.
- **Task C:** Draw the logic circuit diagram for this system.

Question 3: Automated Server Cooling

A data center uses a smart cooling fan system for its server racks. A primary thermostat outputs a 1 if the server temperature is $>80^{\circ}\text{C}$. A secondary ambient room sensor outputs a 1 if the room temperature is $>25^{\circ}\text{C}$. Finally, a maintenance switch outputs a 1 when a technician is actively working on the rack. The cooling fan turns on (output 1) if the server temperature is $>80^{\circ}\text{C}$ and the maintenance switch is NOT active. The fan will also turn on if both the server temperature is $>80^{\circ}\text{C}$ and the ambient room temperature is $>25^{\circ}\text{C}$, even if the maintenance switch is active.

- **Task A:** Infer the logic from the description and create a complete truth table for the cooling fan.
- **Task B:** Write out the Boolean logic expression for the fan.
- **Task C:** Draw the logic circuit diagram for this system.

Question 4: Bank Vault Security Lock

A bank vault requires a strict protocol to open its main electronic lock. The system relies on three inputs: a Manager's Key, a Cashier's Key, and a Time Lock. The Time Lock outputs a 1 if the current time is between 9 AM and 5 PM, and a 0 outside these hours. The Manager's Key and Cashier's Key output 1 when inserted and turned. The vault lock will disengage (output 1) only if the Time Lock is active (1) AND at least one of the two keys (Manager or Cashier) is inserted. If the Time Lock is inactive (0), the vault will strictly remain locked.

- **Task A:** Infer the logic from the description and create a complete truth table for the vault lock.
- **Task B:** Write out the Boolean logic expression for the lock mechanism.
- **Task C:** Draw the logic circuit diagram for this system.

Question 5: Smart Car Defroster

A modern vehicle features an automatic windshield defroster. The system checks an exterior thermometer, which outputs a 1 if the outside temperature is $<4^{\circ}\text{C}$. A cabin humidity sensor outputs a 1 if the interior humidity is $>70\%$. The driver can also press a manual defrost button (outputs 1). The defroster will activate automatically (output 1) if the outside temperature is $<4^{\circ}\text{C}$ AND the humidity is $>70\%$. Additionally, the defroster will immediately activate anytime the manual button is pressed, overriding the automated sensor logic.

- **Task A:** Infer the logic from the description and create a complete truth table for the defroster.
- **Task B:** Write out the Boolean logic expression for the defroster.
- **Task C:** Draw the logic circuit diagram for this system.

Question 6: Factory Conveyor Reject Mechanism

A manufacturing plant uses a sorting mechanism to reject defective packages. The system monitors a weight scale, which outputs a 1 if a package is >5 kg. A dimension scanner outputs a 1 if the package is too tall (>50 cm). A metal detector outputs a 1 if foreign metal is detected inside the box. The reject mechanism (a pneumatic arm) will push a package off the belt (output 1) if foreign metal is detected. It will also reject the package if it is both >5 kg AND too tall (>50 cm).

- **Task A:** Infer the logic from the description and create a complete truth table for the reject mechanism.
- **Task B:** Write out the Boolean logic expression for the pneumatic arm.
- **Task C:** Draw the logic circuit diagram for this system.

Question 7: Elevator Door Safety Sensor

An elevator door uses three safety inputs to determine if it should reopen (output 1) while closing. A laser tripwire outputs a 1 if a person is standing in the doorway. A weight limit sensor outputs a 1 if the elevator is carrying >1000 kg. A mechanical bumper on the door outputs a 1 if it physically strikes an object. The door must reopen immediately if the mechanical bumper strikes an object OR if the laser tripwire is broken. If the weight limit is exceeded (>1000 kg), the door will also reopen, but only if the tripwire is simultaneously broken.

- **Task A:** Infer the logic from the description and create a complete truth table for the elevator door.
- **Task B:** Write out the Boolean logic expression for the door mechanism.
- **Task C:** Draw the logic circuit diagram for this system.

Question 8: Fire Suppression Sprinkler

A commercial kitchen is equipped with an automated fire suppression sprinkler. The system has a smoke detector that outputs a 1 if smoke opacity is $>20\%$. A heat sensor outputs a 1 if the ceiling temperature exceeds $>75^{\circ}\text{C}$. A manual pull station outputs a 1 if pulled by an employee. To prevent false alarms from normal cooking smoke, the sprinkler will only trigger automatically (output 1) if BOTH the smoke opacity is $>20\%$ AND the temperature exceeds 75°C . However, if the manual pull station is activated, the sprinkler will trigger immediately regardless of the environmental sensors.

- **Task A:** Infer the logic from the description and create a complete truth table for the sprinkler system.
- **Task B:** Write out the Boolean logic expression for the sprinkler.
- **Task C:** Draw the logic circuit diagram for this system.

Question 9: Automated Chicken Coop Door

A smart farming setup controls a motorized door for a chicken coop. A light sensor outputs a 1 if ambient sunlight is >500 lux (daytime) and 0 if it is <500 lux (nighttime). An internal motion sensor outputs a 1 if all chickens are inside the coop. A rain sensor outputs a 1 if it is currently raining. The motorized door should close (output 1) if it is nighttime (<500 lux) AND all chickens are inside. The door should also close during the daytime if it starts raining, but only if all chickens are safely inside.

- **Task A:** Infer the logic from the description and create a complete truth table for the motorized door.
- **Task B:** Write out the Boolean logic expression for the door.
- **Task C:** Draw the logic circuit diagram for this system.

Question 10: Drone Battery Saver Mode

An aerial photography drone utilizes a smart battery saver mode to ensure it can return home safely. The system monitors the battery level (outputs 1 if battery is $<15\%$), the distance from the launch pad (outputs 1 if distance is >2 km), and the current wind speed (outputs 1 if wind is >20 km/h). The battery saver mode will activate (output 1) if the battery is $<15\%$ AND the distance is >2 km. It will also activate if the wind is >20 km/h AND the battery is $<15\%$, regardless of the distance, to prevent the drone from fighting strong winds on a low charge.

- **Task A:** Infer the logic from the description and create a complete truth table for the battery saver mode.
- **Task B:** Write out the Boolean logic expression for the battery saver mode.
- **Task C:** Draw the logic circuit diagram for this system.